Major advances in Parkinson's disease over the past two decades and future research directions

Developments in levodopa formulations and the standardisation of deep brain stimulation (DBS) substantially improved clinical management of patients with Parkinson's disease before the turn of the century. As a result of these developments, Parkinson's disease has become a chronic disorder and it is associated with a plethora of non-motor disabling complications. Cognitive impairment is now a major complication of Parkinson's disease.[1] Patients gradually develop from being cognitively normal, to having mild cognitive impairment and then dementia as part of a continuum of progressive pathological changes in the brain, including misfolded protein deposition and neuronal loss.[2] Gait dysfunction, leading to an inability to walk, is another major complication associated with long-term disease progression. Moreover, dysautonomia (eg, urinary dysfunction, constipation, and blood pressure dysregulation) can affect quality of life. These and other manifestations mean that Parkinson's disease is now a chronic multisystem disorder (figure).

Since the discovery of mutations in the synuclein (*SNCA*) and parkin (*PRKN*) genes, several other genes have been linked to risk of Parkinson's disease;[3] however, only *LRRK2, VPS35, PINK1, DJ1*, and *GBA1* have been convincingly associated with the disease. Genetic data show that disease mechanisms converge in common pathways involving dysfunction of lysosomal, mitochondrial, synaptic, and endosomal vesicles and protein recycling. Notably, the monogenic loci recognised to date explain only a small fraction of the familial cases. Some genetic forms (ie, *PRKN*, *PINK1*, and *LRRK2*) underscore the challenge of understanding selective neuronal vulnerability and why a ubiquitous cellular dysfunction is deleterious for a subset of dopaminergic neurons. When a ubiquitous genetic defect affects preferentially the neurons of the substantia nigra, in a process that begins unilaterally, then the gene defect cannot explain, per se, the onset of neurodegeneration.

Neuroimaging has also had extraordinary growth over the past 20 years, with advances in both structural and functional imaging. MRI sequences to assess iron, neuromelanin, and free-water imaging have become promising biomarkers for neurodegeneration,[4] but require confirmatory evidence. Another highly relevant development is that of new PET tracers; serotoninergic and cholinergic systems have been extensively investigated and their contribution to motor and non-motor features and prodromal abnormalities has been identified.[5] Moreover, assessment of dopaminergic terminals in vivo has enabled the identification of a negative exponential progression pattern of dopaminergic loss in the striatum,[6] patterned topographical progression of striatal denervation,[7] and early involvement of axonal terminals. Finally, the use of fluorodeoxyglucose-PET to monitor metabolic changes has been validated as a promising diagnostic and prognostic biomarker.[8] Additionally, initial evidence on amyloid and tau PET has also been reported in the past few years; however, the greatest expectation is for the use of α-synuclein PET tracers, and studies testing the first human α-synuclein binding compounds are ongoing.[9]

Plasma biomarkers are another important area of research, although with few validated results to date. Neurofilament light chain (NfL) and α-synuclein are the most promising biomarkers. NfL is increased in blood and CSF in people with Parkinson's disease, although its sensitivity and specificity for use in individual patients are low. α-Synuclein has been investigated as a diagnostic and prognostic biomarker in several bodily fluids and tissues, although the most consistent results have been obtained in CSF. New methods using α-synuclein seed amplification assays are highly accurate, but their clinical standardisation is still pending.

The past 20 years have witnessed an explosion of data regarding the role of α-synuclein aggregation in cell death, the spreading of the presumed toxic species, and the concept of a peripheral onset of Lewy body pathology. Braak and colleagues suggested a model whereby abnormal synuclein deposition starts in the olfactory tract and the dorsal nucleus of the vagus nerve, and ascends gradually to affect the substantia nigra pars compacta, the amygdala, and the cortex.[10] Furthermore, they found that α-synuclein deposition can also occur in peripheral nerves in the autonomic system (ie, in the gut,

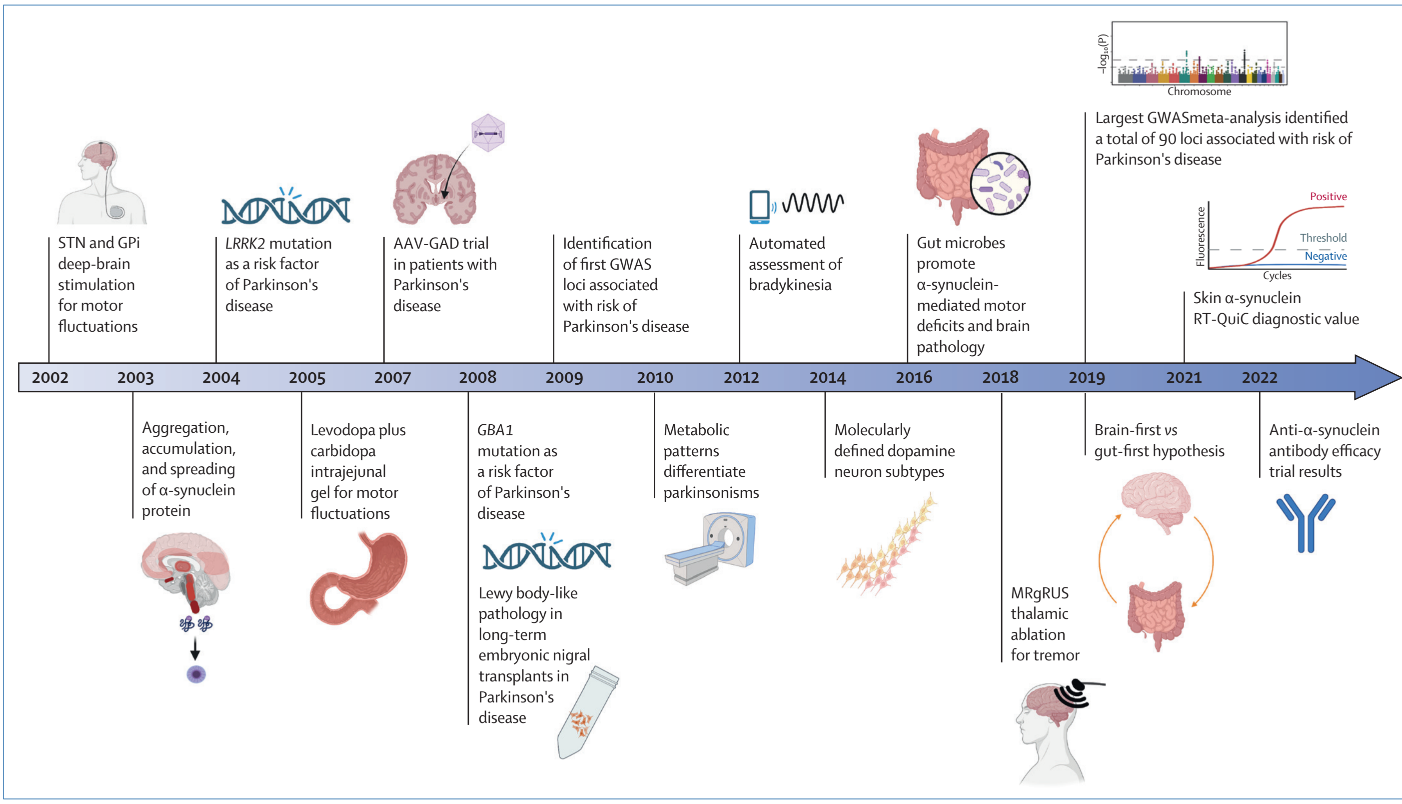

Figure: **Major advances in Parkinson's disease in the past 20 years**
Schematic shows the pathophysiological, diagnostic, and therapeutic advances in Parkinson's disease in the past 20 years. We show the year of regulatory approval for deep brain stimulation, levodopa plus carbidopa intrajejunal, and MRgFUS. AAV-GAD=adeno-associated virus-delivered glutamic acid decarboxylase gene. GWAS=genome-wide association study. GPi=globus pallidus pars interna. MR-gFUS=Magnetic resonance-guided focused ultrasound. RT-QuiC=real-time quaking-induced conversion. STN=subthalamic nucleus.

heart, and skin). The concept caught on rapidly, because it coincided with clinical recognition of some non-motor symptoms (ie, constipation, sleep disorders, and hyposmia), and the Braak staging model was accepted intuitively as the basis for such manifestations.

Subsequently, Lewy body pathology was found in exogenous cells in the striatum of patients several years after transplantation of fetal cells, suggesting that they were "attacked or infected" from the surrounding tissue.[11] This finding in turn led to the prion hypothesis for Parkinson's disease (ie, the spreading of toxic synuclein as the primary mechanism of disease progression). However, Lewy body pathology and inclusions are dissociated from neuronal loss in the substantia nigra pars compacta,[12] the Braak model only fits approximately 50% of patients with Parkinson's disease, and misfolded synuclein (plus several other proteins) propagate from the periphery (eg, the gut) to the brain but also from the brain (striatum) to the gut.[13] The notion of two subtypes of onset has been put forward, suggesting both a peripheral (gut first) and a central (brain first) origin. Notably, this concept is still based on the principle of Lewy body pathology as the key factor leading to neurodegeneration, which is controversial.[14]

A study of individuals aged 60 years or younger with recently diagnosed Parkinson's disease presented a clinical picture dominated by focal cardinal motor features, few and mild non-motor manifestations, and relatively focal dopaminergic loss in the putamen.[15] Disease progression in these patients is mainly determined by nigrostriatal degeneration. Accordingly, early loss of dopaminergic putamen innervation appears to be a fundamental feature in younger patients (ie, aged <60 years) and probably the primary site for pathological events.[14,16] Several dopaminergic neurons can be distinguished in rodents and humans according to their specific gene expression profile (ie, *SOX6* and *ALDH1A1*)[16] that have increased vulnerability and a specific pattern of connectivity.[17]

Histological and neuroimaging studies also support the involvement of neuroinflammation. In the past two decades, several studies of peripheral blood and CSF have shown alterations in inflammation markers and immune cells that could initiate and perpetuate the neurodegenerative process.[18] Inflammation is an area of expanding interest with implications for therapeutic development.

See Online for appendix

Systems to facilitate continuous levodopa delivery (appendix pp 1–3) are now under evaluation in clinical trials and we expect them to be marketed soon. They would facilitate early treatment while optimising anti-parkinsonian effects and reducing the risk of motor complications. Several new drugs have become available (appendix pp 1–3) in the past two decades but none has really made a substantial change in management. The major unmet need is the paucity of therapies that have an effect on disease progression. Notably, the interplay between ageing and neurodegeneration is an unresolved problem that requires increased attention.

The ideal population to test a disease-modifying therapy must be well characterised (eg, mutation carriers) and also relatively young (ie, age ≤60 years) individuals with early Parkinson's disease, restricted clinical manifestations, and focal, nigrostriatal dopaminergic loss.[15] Some of the most relevant approaches tested to date are as follows. First, exercise and a healthy diet are the most efficacious interventions. Both retrospective studies and clinical trials have shown positive effects of aerobic exercise in people with Parkinson's disease, with benefits for both motor and non-motor manifestations.[19] Whether these results are driven by parallel compensation or by slowing neurodegeneration is unknown, but neuroimaging evidence mainly points to increased brain plasticity and enhanced dopamine release.[20] Second, despite initial positive findings after transplantation of striatal fetal dopaminergic cells, enthusiasm for this intervention decreased substantially when a placebo-controlled randomised study did not show significant improvement,[21] and Lewy body pathology was detected in grafted cells. The optimisation of grafts and the timing of implantation are interesting areas in development.[22] Third, early application of DBS has been studied under the hypothesis that earlier restoration of brain activity should lead to slower disease progression, but surgical intervention continues to be considered only in individuals with advanced stages of disease. Focused ultrasound has been used to recreate functional stereotactic approaches. The incisionless and relatively quick and well tolerated nature of the procedure might help in the early treatment of some individuals. Low-intensity focused ultrasound is a new approach to achieve focal blood–brain barrier opening as the first step towards therapeutic delivery of neurorestorative therapies; however, trials are still in development. Finally, in animal models, the administration of neurotrophins can exert vast dopaminergic restorative action. GDNF has been delivered intraventricularly and directly into the putamen of patients without significant clinical results, but with net improvement in dopaminergic striatal activity (as determined by ^{18}F-dihydroxyphenylalanine-PET).[23] Local delivery of gene vectors for enzymatic delivery in the striatum and subthalamic nucleus have been tested but not fully developed.[24]

The field is heading towards early diagnosis and intervention to halt disease progression in individuals with Parkinson's disease. In the foreseeable future, further advances in defining disease subtypes by use of genetics and other biomarkers will probably lead to specific therapies. The availability of neuroimaging biomarkers will allow earlier and accurate diagnosis, and hopefully will resolve the dilemma about the role of Lewy body pathology as a cause or a consequence of neurodegeneration. The development of highly efficacious therapy to stop neurodegeneration will be less easy to achieve but will inevitably occur before *The Lancet Neurology* celebrates its golden jubilee. In the near term, we believe that, by focusing on genetic forms and on individuals with younger age of onset (ie, age <60 years), future research will tackle neurodegeneration in the nigrostriatal system bilaterally and improve quality of life for many years. Admittedly, this aim might not be considered as the ultimate therapeutic goal but is a commendable one if achieved soon.

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*Jose A Obeso, Mariana H G Monje, Michele Matarazzo
jobeso.hmcinac@hmhospitales.com

Centro Integral de Neurociencias AC (CINAC), Hospital Universitario HM Puerta del Sur, Fundación de Investigación HM Hospitales, 28938 Mostoles, Madrid, Spain (JAO, MM); Universidad San Pablo-CEU, Madrid, Spain (JAO); CIBERNED, Instituto de Salud Carlos III, Madrid, Spain (JAO, MM); Ken and Ruth Davee Department of Neurology, Northwestern University, Feinberg School of Medicine, Chicago, IL, USA (MHGM)

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